

FULLERTON, CA, USA

WMO: 722976

Lat: 33.872N

Lon: 117.979W

Elev: 96

StdP: 14.64

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03166

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	40.0	42.8	8.8	8.7	68.1	15.2	11.9	69.9	18.9	66.7	16.3	64.7	1.2	90	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.0	95.3	66.8	91.7	66.8	88.7	66.2	72.4	87.4	71.1	85.3	69.9	83.1	6.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.4	78.5	66.1	96.4	77.3	64.6	91.7	75.5	36.1	87.6	34.9	85.1	33.9	83.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	10.9	10.0	DB	35.2	103.8	2.4	4.3	33.5	106.9	32.1	109.4	30.8	111.9	29.1	115.0
			WB	31.5	74.2	2.1	1.8	29.9	75.5	28.7	76.6	27.5	77.6	25.9	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.1	58.0	58.3	61.2	63.6	66.8	70.5	75.1	76.0	74.4	69.2	62.8	56.9	(6)
	DBStd	8.19	5.15	5.22	5.16	5.23	4.41	3.85	3.92	4.10	5.25	5.14	5.15	4.46	(7)
	HDD50	9	3	2	0	0	0	0	0	0	0	0	0	4	(8)
	HDD65	1042	225	197	139	85	27	2	0	0	0	14	100	253	(9)
	CDD50	5885	250	233	348	409	520	614	777	805	733	596	383	217	(10)
	CDD65	1443	7	8	22	44	82	166	312	340	283	145	33	1	(11)
	CDH74	11384	212	193	320	500	566	903	2131	2540	2191	1272	465	91	(12)
	CDH80	4324	45	58	103	195	216	257	750	1012	959	553	166	10	(13)
(14) Wind	WSAvg	3.7	3.1	3.7	3.9	4.4	4.5	4.4	4.1	3.7	3.4	3.2	2.8	2.9	(14)
(15) Precipitation	PrecAvg	13.3	3.0	3.2	1.8	1.0	0.3	0.1	0.0	0.0	0.2	0.5	1.1	2.0	(15)
	PrecMax	26.4	12.2	13.6	6.2	2.5	2.4	1.1	0.4	0.1	1.7	5.2	3.1	9.8	(16)
	PrecMin	4.3	0.1	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	6.2	3.2	3.0	1.7	0.6	0.5	0.3	0.1	0.0	0.4	1.0	0.8	2.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.0	87.2	90.0	94.1	95.3	93.4	94.9	97.0	100.9	98.1	91.6	81.4	(19)
		MCWB	55.7	56.9	59.8	59.6	60.3	70.1	70.4	70.2	69.8	62.9	59.3	55.3	(20)
	2%	DB	80.4	80.2	81.8	85.3	85.0	86.2	90.4	92.9	94.5	91.3	84.8	76.4	(21)
		MCWB	55.3	56.0	57.7	58.8	61.8	67.7	69.2	69.8	69.5	62.4	57.5	54.1	(22)
	5%	DB	75.1	74.2	76.6	79.3	79.9	82.0	87.3	89.6	90.0	86.2	79.2	72.1	(23)
		MCWB	53.7	55.1	56.7	58.4	63.1	66.5	69.1	69.3	68.1	61.3	56.2	53.0	(24)
	10%	DB	70.3	69.6	72.0	74.3	75.1	79.1	84.0	85.6	85.3	81.0	73.9	67.4	(25)
		MCWB	53.4	54.2	57.0	58.4	61.8	65.5	68.5	68.6	67.2	62.3	56.2	52.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	62.0	64.3	64.7	68.2	70.9	73.8	74.2	74.2	71.2	63.9	61.7	(27)
		MCDB	71.8	73.9	78.2	80.4	82.6	87.6	88.2	90.5	91.7	83.4	75.4	68.6	(28)
	2%	WB	59.3	59.7	61.7	62.4	65.7	68.9	71.7	72.2	72.0	68.2	62.2	59.6	(29)
		MCDB	67.2	69.9	73.8	77.1	78.6	83.3	85.0	87.5	87.9	81.1	73.5	66.2	(30)
	5%	WB	57.7	58.1	60.0	60.9	64.0	67.4	70.5	70.9	70.4	66.2	60.6	57.9	(31)
		MCDB	66.1	67.7	70.9	74.3	76.1	80.4	83.2	85.2	84.6	77.9	71.2	64.5	(32)
	10%	WB	56.3	56.6	58.6	59.3	62.4	66.0	69.2	69.5	69.0	64.5	59.4	56.3	(33)
		MCDB	64.8	65.6	68.1	70.6	72.9	77.5	81.3	82.5	81.8	74.8	68.9	63.1	(34)
(35) Mean Daily Temperature Range	MDBR		22.0	20.1	19.1	18.7	16.3	16.0	17.7	19.0	19.8	20.7	22.1	21.4	(35)
	5% DB	MCDBR	32.0	30.2	29.3	30.1	26.0	22.4	22.5	25.0	27.4	32.4	32.0	30.5	(36)
		MCWBR	13.4	12.8	12.0	11.6	9.9	8.2	7.7	8.6	9.2	11.6	12.5	14.0	(37)
	5% WB	MCDBR	21.0	22.2	20.9	22.3	19.5	20.0	19.8	21.5	22.8	22.0	22.4	19.6	(38)
		MCWBR	10.9	10.9	10.0	9.9	8.4	7.8	7.4	8.2	8.7	9.6	10.6	11.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.332	0.342	0.368	0.371	0.370	0.386	0.389	0.373	0.359	0.363	0.317	0.316	(40)
	taud		2.510	2.492	2.393	2.391	2.428	2.384	2.444	2.475	2.505	2.465	2.593	2.594	(41)
	Ebn at Noon		279	288	288	291	290	284	283	286	287	276	284	279	(42)
	Edh at Noon		27	30	36	38	37	38	36	34	32	31	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1011	1293	1731	2110	2278	2399	2374	2257	1910	1452	1123	916	(44)
	RadStd		71	127	93	105	147	141	123	68	73	102	65	68	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.09	N/A	-4.82	+2.38	N/A	N/A	N/A	-346	+880	+441
(47) Regional (54 neighbors)		+1.12	N/A	-2.97	+1.26	N/A	N/A	N/A	-202	+396	+193

Nomenclature: See separate page